UNIVERSITI MALAYA
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PENILAIAN SUMMATIF ALTERNATIF IJAZAH SARJANA MUDA SAINS KOMPUTER (Bahagian 2)
ALTERNATIVE SUMMATIVE ASSESSMENT FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE (Part 2)

SESI AKADEMIK 2021/2022 : SEMESTER I
ACADEMIC SESSION 2021/2022 : SEMESTER I

WIX1001 : Matematik bagi Pengkomputeran I
Computing Mathematics I

Jan 2022 Masa : 30 minit
Jan 2022 Time : 30 minutes

ARAHAN KEPADA CALON :
INSTRUCTIONS TO CANDIDATES :

Calon dikehendaki menjawab SEMUA SOALAN (10 markah).
Answer ALL QUESTIONS (10 marks).

Ini adalah penilaian buku terbuka dalam talian.
This is an online open book assessment.

Penggunaan kalkulator dibenarkan dalam peperiksaan ini.
The use of calculators are permitted in this examination.

(Kertas soalan ini mengandungi 1 soalan dalam 2 halaman yang dicetak)
(This question paper consists of 1 questions on 2 printed pages)

Question 1

Given the relations R_1 and R_2 as the following:

$R_1 = \{(1, x), (2, x), (2, y), (3, y)\}$
$R_2 = \{(x, a), (x, b), (y, a), (y, c)\}$

a) Find the matrix A_1 of the relation R_1 relative to the orderings*:
1, 2, 3; x, y

b) Find the matrix A_2 of the relation R_2 relative to the orderings*:
x, y; a, b, c

c) Find the matrix product A_1A_2.

d) Use the result obtain in c) to find the matrix in the relation $R_2 \circ R_1$.

e) From your answer in d), determine if the relation is transitive.

* Entry in row x and column y is set to 1 if xRy and to 0 otherwise and this is known as matrix of relation.

END

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PENILAIAN SUMMATIF ALTERNATIF IJAZAH SARJANA MUDA SAINS KOMPUTER (Bahagian 3)
ALTERNATIVE SUMMATIVE ASSESSMENT FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE (Part 3)

SESI AKADEMIK 2021/2022 : SEMESTER I
ACADEMIC SESSION 2021/2022 : SEMESTER I

WIX1001 : Matematik bagi Pengkomputeran I
Computing Mathematics I

Jan 2022 Masa : 60 minit
Jan 2022 *Time : 60 minutes*

ARAHAN KEPADA CALON :
INSTRUCTIONS TO CANDIDATES :

Calon dikehendaki menjawab SEMUA SOALAN (20 markah).
Answer ALL QUESTIONS (20 marks).

Ini adalah penilaian buku terbuka dalam talian.
This is an online open book assessment.

Penggunaan kalkulator dibenarkan dalam peperiksaan ini.
The use of calculators are permitted in this examination.

(Kertas soalan ini mengandungi 3 soalan dalam 3 halaman yang dicetak)
(This question paper consists of 3 questions on 3 printed pages)

Question 1

a) Out of 8 consonants and 5 vowels, how many different 6 letters words of 4 consonants and 2 vowels can be formed?

(2 marks)

b) How many distinct ways can the alphabets of the word 'CORPORATIONS' be arranged, with the condition that the vowels in the word are always together?

(3 marks)

c) In an internal telephone system of a company, a phone number has 6 digits and does not start with either 0 or 1. How many odd telephone number can be assigned in the system?

(2 marks)

d) According to Chinese zodiac, there are 12 signs, including rat, ox, tiger and etc. Everyone belongs to one, and only one sign. In a village, there are at least 8 residents having the same sign, for each of the signs in the zodiac system. Compute at least how many residents in the village. State the name of the theory/principle you used to determine this answer.

(3 marks)

Question 2

The year is 2140 and the space technology of human civilization has advanced greatly. You work at MASA (Malaysia Aeronautics and Space Administration) and need to arrange a group of 8 satellites surrounding a mothership for a mission to the planet Jupiter. Each satellite has exactly one identification code, but, an identification code does not belong to only one satellite. Once an identity code is assigned to a satellite, another 3 satellites receive the same identity code. However, due to Elon Theorem (an ancient theorem established circa 2040), each satellite must be linked to all satellites of different identification code.

a) Given that the vertices are the satellites and links between satellites are edges, draw the complete graph of this relation by indicating the mothership's location. Define all of the graph's components such as the set of vertices $v \in V$ and the set of edges $e \in E$, $G=(V, E)$. State the type of the graph and give your reason.

(3 marks)

b) How many satellites are linked to each satellite? What is the total number of links? If the satellites are required to link to all other satellites, regardless of their identification code, compute the difference in the links required compared to the previous case.

(2 marks)

c) Do you think an Euler cycle exists in the graph G in part (a)? Give reasons for your answer. Given that it costs 0.5 energy to pass a message from one satellite to another linked satellite, compute the energy cost for a message originating in a satellite x to pass through all other satellites and returning to x.

(1 mark)

Question 3

V is a set defined as $V=\{a \in \mathbb{N} : 1 \leq a \leq 15\}$. Here we assume that the elements of set V are vertices. There is an edge from vertex α to vertex β ($\alpha, \beta \in V$) when $\alpha R \beta$ holds. $\alpha R \beta$ is defined as follow:

$\alpha R \beta$ if ($\beta = 2\alpha$) or ($\beta = 2\alpha + 1$)

Draw the graph for this relation. What kind of graph is this?

(4 marks)

END

WIX1001 Computational Mathematics 1
2021 / 2022 Semester 1
Test 1

General Instructions:
a) There are 30 questions in the test
b) Each question carries 1 mark.
c) Choose the best answer for each question.
d) Submit your answers to : https://forms.gle/djZSFsqvgTCwcMdUA
e) Click the "SUBMIT" button before the end of the test, otherwise your answers will NOT be recorded.
f) After " SUBMIT" button is clicked, you are still allowed to click on "Edit your response" to revise your answers.
g) Please submit early or on time. Marks will be deducted for every minute late.

1. Given that $P=\{10\}$, $Q=\{5\}$, $R=\{15\}$. Which of the following is true?
   a) $\{P\}=10$, and $\{Q\}=5$
   b) $\{P\}=\{\{10\}\}$, and $R=15$
   c) 10 is an element of P, and $\{15\}$ is a subset of R
   d) Q is a subset of P, and P is a subset of R

2. Given that $P=\{10\}$, $Q=\{5\}$, $R=\{15\}$. Which of the following is true?
   a) $P \cup Q = R$
   b) $n(P \cup P) > n(Q \cap R)$
   c) $P^c = Q$
   d) $R - Q = P$

For question 3 to 5, refer to the following:

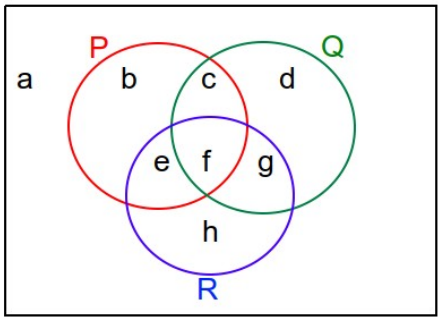
In a small village of 66 villagers, 32 of them are male and the rest are female. With a survey, we find out that only 20 of these villegers like chocolates, and out of these chocolate lovers, 8 of them are male.

3. The number of female villegeres who like chocolates.
   a) 8
   b) 12
   c) 22
   d) 24

4. The number of male villegeres who don't like chocolates.
   a) 8
   b) 12
   c) 22
   d) 24

5. The number of female villegeres who don't like chocolates.
   a) 8
   b) 12
   c) 22
   d) 24

6. Refer to the following figure. Which of the area is represented by $(P\text{-}Q)\cap R$?



a) e
b) f
c) g
d) h

For question 7 to 10, refer to the following:

Given that A = { 1, 2 } and B = { 1, 2, 3 }.

7. Which of the following is true?
   a) $\mathcal{P}$(A) = (1,2)
   b) $\mathcal{P}$(A) = { {1,}, {2} }
   c) $\mathcal{P}$(A) = { {1}, {2}, {1,2} }
   d) A ∈ $\mathcal{P}$(A)

8. n($\mathcal{P}$(B)) – n($\mathcal{P}$(A)) = ____
   a) 2
   b) 4
   c) 8
   d) 16

9. A X B =
   a) 6
   b) {1, 2, 3, 2, 4, 6 }
   c) 1 + 2 + 3 + 2 + 4 + 6 =18
   d) { (1,1), (1,2), (1,3), (2,1), (2,2), (2,3) }

10. If p ∈ AxB and q ∈ BXA, _____ .
    a) There is a chance that p = q
    b) p is always not q
    c) p ∈ A and q ∈ B
    d) n(AxB) > n(BxA)

11. Given that P is the set of prime numbers, and $M = \{(x+1)^2 \mid x \in P\}$. Which of the follwoing statement is FALSE?
    a) If $y \in P$, then $y \notin P$'
    b) If $y \in P$, then $y \notin M$
    c) $16 \in M$
    d) $11 \in M$

12. Which of the following is a proposition?
    a) If I am late for work, would you help me to prepare breakfast?
    b) Saloma drinks coffee or tea for breakfast.
    c) Bring me coffee as my breakfast.
    d) You prefer green tea or black tea?

13. Which of the following is NOT a proposition?
    a) Mathematics is very easy.
    b) Shout “open sesame” to open the door.
    c) The door opened when you say “open sesame.”
    d) There is only a bag of sesame behind the mysterious door.

14. Which of the following has a FALSE truth value?
    a) odd numbers become even if multiply by 2 and even numbers become odd if divided by 2
    b) If Malaysia is not a South East Asia country then USA is a Middle East country.
    c) It is not the case that I am not attending a mathematics course in the University of Malaya.
    d) $5 - 3 \geq 2$ or mathematics has no relation with computer science

15. “If James Bond has quit his job, then I will become James Bond’s successor”
    This statement is NOT equivalent to:
    a) I will become James Bond’s successor if James Bond has quit his job
    b) I will become James Bond’s successor is sufficient for him to quit his job.
    c) If James Bond has quit his job, I will become his successor
    d) I will become James Bond’s successor when he quit his job

16. “if you have a pen, then you can write the answer.”
    The contrapositive of this proposition is:
    a) If you cannot write the answer, then you do not have a pen.
    b) If you can write the answer, you have a pen.
    c) If you do not have a pen, you cannot write the answer.
    d) For you to write the answer, you do not need a pen

17. $(P \vee Q) \leftrightarrow (Q \vee P)$ is a _____ .
    a) tautology
    b) contradiction
    c) contigency
    d) contrapositive

18. Refer to the following argument:

When I drink coffee, I eat biscuits I am not eating biscuit Therefore, I am not drinking coffee

This argument is _____.

a) Valid because of modus ponens

b) Valid because of modus tollens

c) Valid because of some other rules of inferences besides modus ponens and modus tollens

d) Invalid

19. Refer to the following argument:

All my enamies wear black shirts John Doe wear black shirt Therefore, John Doe is my enamy

This argument is _____.

a) Valid because of modus ponens

b) Valid because of modus tollens

c) Valid because of some other rules of inferences besides modus ponens and modus tollens

d) Invalid

20. Refer to the following argument:

Black Widow and Iron Man will become a Malaysian one day Iron Man will not become a Malaysian one day Therefore, Black Widow will become a Malaysian one day

This argument is _____.

a) Valid because of modus ponens

b) Valid because of modus tollens

c) Valid because of some other rules of inferences besides modus ponens and modus tollens

d) Invalid

21. Refer to the following argument:

If Mr Bean has Mrs Bean, then Marry Poppins can fly Mr Bean does not have Mrs Bean Therefore, Marry Poppins cannot fly

This argument is _____.

a) Valid because of modus ponens

b) Valid because of modus tollens

c) Valid because of some other rules of inferences besides modus ponens and modus tollens

d) Invalid

22. Symbol ∃x represent _____ .

a) For all x
b) For each x
c) Some of the x
d) Only one x

23. If A(x) represents “x can fly” and the domain of discourse of x is birds. Which of the following is true?

a) ∀xA(x) represents “some birds can fly”
b) ∀xA(x) represents “all birds can fly”
c) ∃xA(x) represents “all birds can fly”
d) ∃xA(x) represents “no birds can fly”

24. If $A(x)$ is “x reads poetries” and x refers to all UM students.

a) $\forall x\, A(x)$ represents all students read poetries.
b) $\exists x\, A(x)$ represents some UM students do not read poetries.
c) $\forall x\, A(x)$ represents all UM students read poetries
d) $\exists x\, A(x)$ represents some UM students love musics

25. Let $x \in \mathbb{Z}$ and B(x) represents $x^2>0$. Pick a FALSE statement.

a) ∃x B(x) is false
b) ∃x B(x) is true
c) ∀x B(x) is true
d) ∀x B(x) is false

26. Let x and y be natural numbers and C(x,y) represents $\frac{x}{y} \in \mathbb{Z}$. Pick a TRUE statement.

a) ∀x∀y $C(x,y)$ is true
b) ∃x∀y $C(x,y)$ is false
c) ∀x∃y $C(x,y)$ is true
d) ∃x∃y $C(x,y)$ is true

27. The negation of “All cards in this box have a squared shape” is _____ .

a) All cards in this box have a rounded corner
b) All cards in this box have a rounded shape
c) None of the cards in this box have a squared shape.
d) There is at least a card in this box has a non-squared shape.

28. ~∃xP(x) is equivalent to _____.

a) ∀x~P(x)
b) ~∀xP(x)
c) ∃x~P(x)
d) ∃xP(x)

29. “There are some authors that like to print all thier books in black and white only.” If x refers to the authors and y refers to the printing of thier books, the this quantified proposition can be written in the form:

a) $\forall x \forall y\ B(x,y)$

b) $\exists x \exists y\ B(x,y)$

c) $\exists x \forall y\ B(x,y)$

d) $\forall x \exists y\ B(x,y)$

30. Given that P(x) : “x likes movies” and Q(x) : “x is happy”. $\forall x(P(x) \rightarrow Q(x))$ represents _____.

a) Those who like movies are happy

b) There is an x who likes movies and happy

c) Some people who likes movies are happy

d) Everyone likes movies and happy.

End

WIX1001 Computational Mathematics 1
2021 / 2022 Semester 1
Test 2

General Instructions:
a) There are 25 questions in the test
b) Each question carries 1 mark.
c) Choose the best answer for each question.
d) Submit your answers to : https://forms.gle/XN3Evd8papWwiS9T8
e) Click the "SUBMIT" button before the end of the test, otherwise your answers will NOT be recorded.
f) After " SUBMIT" button is clicked, you are still allowed to click on "Edit your response" to revise your answers.
g) Please submit early or on time. Marks will be deducted for every minute late.

For question 1 – 4, refer to the following definitions

w is a row vector with 5 elements **x** is a column vector with 5 elements

1. The result of **wx** is _____ .
   a) A square matrix **M** with dimension 5
   b) A column vector
   c) A column vector
   d) *A scalar*

2. Which of the following operation produce a 5x5 matrix?
   a) $\mathbf{w}\mathbf{w}^T$
   b) $\mathbf{w}^T\mathbf{w}$
   c) $\mathbf{w}\mathbf{w}^T\mathbf{w}$
   d) $\mathbf{w}^T\mathbf{w}\mathbf{w}^T$

3. Which of the following operation produce a column vector?
   a) $\mathbf{w}\mathbf{w}^T$
   b) $\mathbf{w}^T\mathbf{w}$
   c) $\mathbf{w}\mathbf{w}^T\mathbf{w}$
   d) $\mathbf{w}^T\mathbf{w}\mathbf{w}^T$

4. Which of the following operation produce a row vector?
   a) $\mathbf{w}\mathbf{w}^T$
   b) $\mathbf{w}^T\mathbf{w}$
   c) $\mathbf{w}\mathbf{w}^T\mathbf{w}$
   d) $\mathbf{w}^T\mathbf{w}\mathbf{w}^T$

5. If **A** is a symmetric matrix, which operation is equivalent to $\mathbf{A}^2$?
   a) $\mathbf{A}\mathbf{A}^{-1}$
   b) $(\mathbf{A}^T\mathbf{A})^T$
   c) $\mathbf{A}^{-1}\mathbf{A}$
   d) $|\mathbf{A}|\,|\mathbf{A}|$

6. Given **A** is a *m* x *n* matrix with $m \neq n$, which operation is possible?
   a) $\det(\mathbf{A}\mathbf{A}^T)$
   b) $\det(\mathbf{A})$
   c) $\mathbf{A}^{-1}$
   d) $\det(\mathbf{A}^{-1})$

7. Given **D** is a *n* x *n* diagonal matrix such that
$$\boldsymbol{D} = \begin{bmatrix} 1 & 0 & \cdots & \cdots & \cdots & 0 \\ 0 & 2 & 0 & & & \vdots \\ \vdots & 0 & 3 & & & \vdots \\ \vdots & & & \ddots & & \vdots \\ \vdots & & \cdots & & n-1 & \vdots \\ 0 & \cdots & \cdots & \cdots & \cdots & n \end{bmatrix}$$
   What is $|\mathbf{D}|$?
   a) $2n^2$
   b) $(n+2n)/3$
   c) $n!$
   d) $n!/2n$

8. How many bit strings of length 10 either begin with three 0s or end with two 0s, but not both?
   a) 384
   b) 352
   c) 128
   d) 256

9. A student can choose a computer project from one of the three lists. The three lists contain 23, 15, and 19 projects respectively. No project is on more than one list. How many possible projects are there to choose from?
   a) P(57,1)
   b) P(57,23)
   c) P(57,15)
   d) P(57,19)

10. There are 10 distinct cards for a lucky draw competition, with only 4 of them are lucky cards with prizes. You draw 3 cards. How many ways are there when at least one of them is a lucky card?
   a) 100
   b) 80
   c) 55
   d) 120

11. There are 55 students in a class. Every student sends a single greeting card to all of the other students. In total, how many greeting cards were sent by students in that class?
   a) 990
   b) 890
   c) 2970
   d) 1980

12. In how many ways can we select 7 people out of 11, of which a particular person is not included?
   a) C(11,4)
   b) C(11,7)
   c) C(10,7)
   d) C(10,5)

13. There are 100 students in WIX1001 celebrate their birthdays in January 2022. Therefore, at least x students of WIX1001 celebrate their birthdays on the same day of January 2002.

a) $x \leq 3$
c) $x = 5$
b) $x = 4$
d) $x \geq 6$

14. There are 20 distinct cards. We draw 5 cards and record all the cards we have drawn. After replacing all these cards, we repeat the same action again. In how many ways can this be done?

a) C(20,5)
c) 2 x C(20,5)
b) C(20,10)
d) C(20,5) x C(20,5)

15. How many strings of eight English letters are there that contains no vowels, if letters can be repeated?

a) 390,625
c) 37,822,859,361
b) 16,777,216
d) 2,520

16. Which of the following is the correct steps to take when proving a statement using proof by contradiction?

a) 1. Assume that your statement is true.
2. Show this is the case using definitions and theorems.
3. State that the statement is true.

b) 1. Assume your statement is true for a certain instance.
2. Show that it is true in more than one instance.
3. State that your statement must be true.

c) 1. Assume your statement to be false.
2. Proceed as you would in a direct proof.
3. Come across a contradiction.
4. Use the contradiction to state that your assumption of the statement being false can't be the case, so your statement must be true.

d) None of the answers are correct.

17. We aim to prove $\sqrt{2}$ as irrational. We start with an assumption that $\sqrt{2}$ is rational. This relates to which type of proof?

a) Direct proof
c) Disproving with counter example
b) Proof by Contradiction
d) Proof by cases

18. A proof is achieved by breaking it up into distinct cases. These cases cover all prospects. These kinds of proofs are known as _____ .

a) Direct proof
c) Proof by Contradiction
b) Contrapositive proofs
d) Proof by cases

19. Let the statement be "If n is not an odd integer, then square of n is not odd.".
$P(n)$ is then defined as "n is an not an odd integer", and $Q(n)$ is defined as "(square of n) is not odd."
For direct proof we should prove which of the followings:

a) $\forall n\ P((n) \rightarrow Q(n))$
c) $\forall n{\sim}(P((n)) \rightarrow Q(n))$
b) $\exists\ nP((n) \rightarrow Q(n))$
d) $\forall nP((n) \rightarrow {\sim}(Q(n)))$

20. Let the statement be “If n is not an odd integer, then sum of n with some not odd number will not be odd.”. P(n) is then defined as “n is an not an odd integer” and Q(n) is defined as “sum of n with some not odd number will not be odd.” A proof by contrapositive is which of the followings:

a) $\forall$nP ((n) → Q(n))
b) $\exists$ nP ((n) → Q(n))
c) $\forall$n~(P ((n)) → Q(n))
d) $\forall$n(~Q ((n)) → ~(P(n)))

21. How many one-to-one functions are there from a set of five elements to sets of five elements?

a) 25
b) 120
c) 3125
d) 125

22. Given relation R = {(1,1), (1,4), (2,3), (3,1), (3,4)} and S = {(1,0), (2,0), (3,1), (3,2), (4,1)}. Which of these relations is NOT in the composition SoR?

a) (1,0)
b) (2,2)
c) (3,1)
d) (3,4)

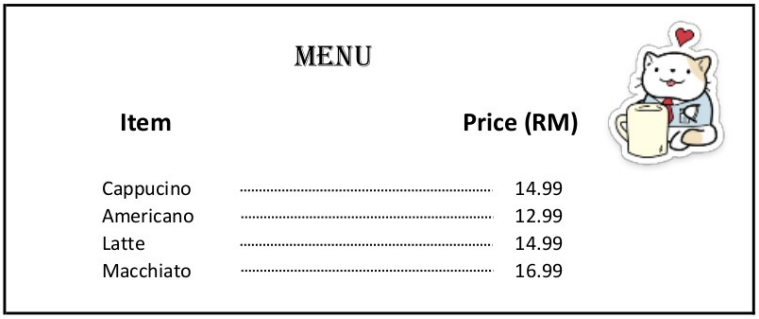
23. A relation R = {(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)} is defined on a set containing {1, 2, 3, 4}. Determine the properties of these relation.

a) reflexive, and antisymmetric
b) symmetric
c) reflexive, symmetric and transitive
d) None of the above

24. What is the inverse function of f(x) = 3(x − 2)^3?

a) $f^{-1}(x) = \left(\frac{x}{3}\right)^{\frac{1}{3}} + 2$
b) $f^{-1}(x) = \frac{x^{\frac{1}{3}}}{3} + 2$
c) $f^{-1}(x) = \frac{1}{3}(x+2)^{\frac{1}{3}}$
d) $f^{-1}(x) = 3x^{\frac{1}{3}}$

25. Given the following menu list. Determine the TRUE statement:



a) The price is not a function of the item.
b) The item is a function of the price.
c) It is a one-to-many relation.
d) This is a function if x = Item and y = Price (RM)

End

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PEPERIKSAAN IJAZAH SARJANA MUDA SAINS KOMPUTER / SARJANA MUDA TEKNOLOGI MAKLUMAT
EXAMINATION FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE / BACHELOR OF INFORMATION TECHNOLOGY

SESI AKADEMIK 2015/2016 : SEMESTER I
ACADEMIC SESSION 2015/2016 : SEMESTER I

WIX1001 : Matematik Pengkomputeran I
Computing Mathematics I

Dis 2015/Jan 2016 Masa: 2 jam
Dec 2015/Jan 2016 Time: 2 hours

ARAHAN KEPADA CALON:
INSTRUCTIONS TO CANDIDATES:

Calon dikehendaki menjawab **LIMA (5)** soalan sahaja (50 markah).
*Answer **FIVE (5)** questions only (50 marks).*

(Kertas soalan ini mengandungi 5 soalan dalam 7 halaman yang dicetak)
(This question paper consists of 5 questions on 7 printed pages)

1. a) Diberi set

Given set

$$X = \{2, 3, 5, 7, 9\}$$

dan

and

$$R = \{(x, y) \mid (x \leq y\}$$

satu hubungan di atas set X.

a relation on the set X.

Senaraikan semua elemen R dan tentukan sama ada R adalah refleksif, simetrik, dan/atau transitif. Adakah R suatu hubungan setara.

List all the elements of R and determine whether R is reflexive, symmetric and/or transitive. Is R an equivalence relation.

(5 markah/*marks*)

b) Diberi fungsi-fungsi:

Given the functions:

$$f(x) = |2x - 3|\ ,\ g(x) = \frac{x}{2}$$

dan

and

$$h(x) = x^2 - 5.$$

i) Dapatkan yang berikut, kemudian nyatakan domain dan julatnya.

Find the following, then state its domain and range.

$$F(x) = (f \circ g \circ h)(x)$$

ii) Lakar dan labelkan graf bagi fungsi berikut.

Sketch and label the graph for the following function.

$$F(x)$$

iii) Apakah nilai bagi yang berikut?

What is the value of the following?

$$F(0)$$

(5 markah/*marks*)

2. a) Biar hubungan

Let relation

$$g: R \rightarrow R$$

ditakrifkan sebagai

be defined as

$$g(x) = -5x^2 + 10$$

Tentukan sama ada $g(x)$ adalah injektif, surjektif dan/atau bijektif.

Determine whether $g(x)$ is injective, surjective and/or bijective.

(3 markah/*marks*)

b) Dapatkan songsangan bagi fungsi berikut:

Find the inverse for the following function:

$$f(x) = 4x^2 - 4x$$

(2 markah/*marks*)

c) Dengan menggunakan Jadual Kebenaran, tunjukkan yang berikut. Seterusnya, nyatakan sama ada ianya suatu Tautologi atau tidak.

Using the Truth Table, show the following. Consequently, state if it is a Tautology or not.

$$(p \vee q) \wedge \sim(\sim p \wedge r)$$

(5 markah/*marks*)

3. a) Diberi

Given

$$n(U) = 150,\ n(A) = 37,\ n(B) = 51,\ n(C) = 42$$

Jika persatuan A, B dan C adalah 120, persilangan A, B dan C adalah 8, dan kesemua persilangan A dan B, A dan C, dan B dan C mempunyai nilai yang sama,

If the union of A, B and C is 120, the intersection of A, B and C is 8, and the intersections between A and B, A and C, and B and C are all of equal values,

i) Cari bilangan elemen yang dipunyai oleh A sahaja, B sahaja dan C sahaja?

Find how many elements belong to A only, B only and C only?

ii) Lakarkan rajah Venn bagi komposisi U dan isikan rajah dengan maklumat yang bersesuaian.

Draw a Venn diagram representing the composition of U and fill the diagram with appropriate information.

iii) Apakah peratusan elemen-elemen yang berada di luar A, B dan C berbanding U?

What is the percentage of elements outside of A, B and C compared to U?

(4 markah/*marks*)

b) Dengan menggunakan Hukum-Hukum Logik (rujuk Lampiran), buktikan yang berikut dengan menunjukkan persamaan yang di sebelah kiri setara dengan persamaan di sebelah kanan.

Using the Laws of Logic (refer Attachment), proof the following by showing the left hand side statement is equivalence to the right hand side statement.

i) $p \rightarrow (q \wedge r) \equiv (p \rightarrow q) \wedge (p \rightarrow r)$

ii) $(p \rightarrow q) \rightarrow (q \rightarrow p) \equiv q \rightarrow p$

iii) $\sim(\sim p \vee (p \vee q)) \rightarrow q \equiv r \vee \sim r$

(6 markah/*marks*)

4. a) i) Berapakah bilangan kata laluan 6-digit yang boleh dicipta dengan menggunakan digit 0-5?

How many 6-digit passwords can be created using digits 0-5?

ii) Berapakah bilangan kata laluan 6-digit yang boleh dicipta dengan menggunakan digit 0-5, jika "0" tidak boleh digunakan sebagai digit pertama dan terakhir dan kesemua digit-digit tidak boleh berulang?

How many 6-digit passwords can be created using digits 0-5, if the first and last digit cannot be "0" and the digits cannot be repeated?

(3 markah/*marks*)

b) Andaikan sekumpulan pelajar A, B, C, D dan E dan mereka mempunyai hubungan-hubungan seperti berikut:

Suppose there is a group of students A, B, C, D *and* E *and they have the following relations:*

$$A - C, A - D, B - C, C - D, C - E$$

i) Lakarkan graf G untuk mewakili situasi ini.

Draw a graph G to represent this situation.

ii) Dengan notasi set, senaraikan semua bucu (V) dan tepi (E).

Using the set notation, list all vertex (V) and edges (E).

iii) Lakarkan matriks bersebelahan untuk G.

Draw an adjacency matrix for G.

(5 markah/*marks*)

c) Diberi

Given

$$A = \{\emptyset, \{a\}\}$$

Tentukan sama ada pernyataan berikut betul atau salah.

State if the following statement is true or false.

i) $\emptyset \in A$
ii) $\{\{a\}\} \subseteq \{A\}$
iii) $\{\emptyset, \{a\}\} \subset A$
iv) $\{A\}$ has 4 subsets

(2 markah/*marks*)

5. a) i) Diberi

Given

$$A = \begin{bmatrix} 3 & -7 \\ 5 & 8 \end{bmatrix}$$

Jika

If

$$A^2 + pA + qI = I^2$$

di mana p dan q adalah nyata, dan I adalah matriks identity,

where p and q are real, and I is the identity matrix,

cari

find

p

dan

and

q

ii) Selesaikan sistem persamaan linear yang berikut menggunakan kaedah matriks sonsang.

Solve the following system of linear equations by the inverse matrix method.

$$\begin{aligned} x + 2y + 3z &= 5 \\ 7x + y + 6z &= 3 \\ -y + 2z &= 7 \end{aligned}$$

(8 markah/*marks*)

b) Suatu kotak mengandungi 2 biji bola putih, 3 biji bola hitam dan 4 biji bola merah. Berapakah cara 3 biji bola boleh dipilih dari kotak tersebut, jika sekurang-kurangnya sebiji bola hitam mesti termasuk dalam pilihan?

A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box, if at least one black ball is to be included in the draw?

(2 markah/*marks*)

Lampiran
Attachment

Hukum Logik
The Laws of Logic

1	$\neg\neg p = p$	Double Negation
2	$p \wedge q = q \wedge p$	Commutative Law
3	$p \vee q = q \vee p$	Commutative Law
4	$(p \wedge q) \wedge r = p \wedge (q \wedge r)$	Associative Law
5	$(p \vee q) \vee r = p \vee (q \vee r)$	Associative Law
6	$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	Distributive Law
7	$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$	Distributive Law
8	$p \wedge p = p$	Idempotent Law
9	$p = p$	Idempotent Law
10	$p \wedge T = p$	Properties of TRUE (T)
11	$p \vee T = T$	Properties of TRUE (T)
12	$p \vee F = p$	Properties of FALSE (F)
13	$p \wedge F = F$	Properties of FALSE (T)
14	$p \vee p = p$	Identity Law
15	$p \wedge p = p$	Identity Law
16	$p \vee \neg p = T$	Inverse Law
17	$p \wedge \neg p = F$	Inverse Law
18	$p \rightarrow q = \neg p \vee q$	Law of Implication
19	$p \wedge (p \vee q) = p$	Absorption Law
20	$p \vee (p \wedge q) = p$	Absorption Law
21	$\neg(p \vee q) = \neg p \wedge \neg q$	De Morgan's Law

TAMAT
END

UNIVERSITI MALAYA
UNIVERSITY OF MALAYA

PEPERIKSAAN IJAZAH SARJANA MUDA SAINS KOMPUTER / SARJANA MUDA TEKNOLOGI MAKLUMAT
EXAMINATION FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE / BACHELOR OF INFORMATION TECHNOLOGY

SESI AKADEMIK 2016/2017 : SEMESTER I
ACADEMIC SESSION 2016/2017 : SEMESTER I

WIX1001 : Matematik Pengkomputeran I
Computing Mathematics I

Dis 2016/Jan 2017
Dec 2016/Jan 2017

Masa: 2 jam
Time: 2 hours

ARAHAN KEPADA CALON:
INSTRUCTIONS TO CANDIDATES:

Calon dikehendaki menjawab **LIMA (5)** soalan (100 markah).
*Answer **LIMA (5)**questions (100 marks).*

(Kertas soalan ini mengandungi 6 soalan dalam 8 halaman yang dicetak)
(This question paper consists of 6 questions on 8 printed pages)

1. a) A, B dan C merupakan set-set dalam set sejagat U. Lakarkan gambar rajah Venn yang sesuai untuk menggambarkan setiap yang berikut:

Let A, B and C be sets in the universe U. Draw appropriate Venn Diagrams to illustrate each of the following:

i) $A \cup B$

ii) $B \cap C$

iii) $(A - B)$

iv) $(A \cap B) \cup (B \cap C)$

v) $(A - B) \cup (B - A)$

(10 markah/*marks*)

b) Biar $A \Theta B = (A - B) \cup (B - A)$ atau juga dikenali sebagai perbezaan simetrik, bagi A dan B satu set yang mengandungi unsur-unsur di dalam A atau B, tetapi bukan kedua-duanya. Sekiranya

Let $A \Theta B = (A - B) \cup (B - A)$ also known as the symmetric difference of A and B, be the set containing those elements in either A or B, but not both A and B. If

$A = \{-2, 0, 2, 4, 6, 8\}$, $B = \{-2, -1, 0, 1, 2, 3, 4\}$

dan

and

$C = \{2, 3, 4, 5, 6, 7, 8\}$.

Dapatkan

Find

i) $A \Theta B$

ii) $A \Theta C$

iii) $B \Theta C$

iv) $(A \Theta B) \Theta C$

(10 markah/*marks*)

2. a) R merupakan satu hubungan dalam set integer $A = \{0, 1, 2, 3, 4, 5\}$ yang ditakrif sebagai $R = \{(x, y)$: x dan y mempunyai baki yang sama bila dibahagikan dengan 5$\}$. Nyatakan pasangan bertertib yang ditakrif oleh A dan hubungan R yang mempunyai baki

R is a relation on the set of integers on $A = \{0, 1, 2, 3, 4, 5\}$ defined as $R = \{(x, y)$: x and y have the same remainder when divided by 5$\}$. State the ordered pairs defined on the set A and the relation R that has a remainder of

i) 0

ii) 1

iii) 2

iv) 3

v) 4

vi) Adakan R satu hubungan setara?
Is R an equivalence relation?

(10 markah/*marks*)

b) Bagi setiap fungsi berikut f: R → R, tentukan samada fungsi adalah sujektif, injektif atau/dan bijektif. Sekiranya fungsi bijektif, dapatkan fungsi songsangannya.

For each of the following functions f: R → R, decide whether the function is injective, surjective, or/and bijective. If the function is bijective, find its inverse.

i) $f(x) = \ln(x),\ x > 1$

ii) $f(x) = {}^{x}/_{5} - 1$

iii) $f(x) = \log(x) - 1$

(10 markah/*marks*)

3. a) Tuliskan proposisi berikut dalam notasi logik simbolik dan berikan jadual kebenaran bagi setiap satu proposisi tersebut.

Write the propositions below in the symbolic notation of propositional logic and provide their respective truth tables.

i) Hari ini bukan hari Isnin.

Today is not Monday.

ii) Hari ini hari Isnin dan tak hujan.

Today is Monday and it is not raining.

iii) Hari ini bukan hari Isnin dan tak hujan.

Today is not Monday and it is not raining.

iv) Samada hari ini hari Isnin atau ianya hujan.

Either today is Monday or it is raining.

v) Samada hari ini hari Isnin atau ianya tak hujan.

Either today is Monday or it is not raining, but today is Monday.

(10 markah/*marks*)

b) Dengan mengguna Hukum Mantik yang diberi dalam Jadual 1, tentukan samada proposisi berikut adalah setara secara logik.

Using the Laws of Logic provided in Table 1, determine if the following propositions are logically equivalent.

i) p dan/and $\neg((\neg p \wedge (\neg p \vee q)) \wedge (\neg p \vee (\neg p \wedge q)))$

ii) $p \wedge \neg q$ dan/and $\neg(p \vee \neg q) \vee (p \rightarrow q)$

iii) $(p \rightarrow q) \vee q$ dan/and $q \vee (\neg p \vee q)$

(10 markah/*marks*)

$\neg\neg p = p$	Double Negation
$p \wedge q = q \wedge p$	Commutative Law
$p \vee q = q \vee p$	Commutative Law
$(p \wedge q) \wedge r = p \wedge (q \wedge r)$	Associative Law
$(p \vee q) \vee r = p \vee (q \vee r)$	Associative Law
$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	Distributive Law
$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$	Distributive Law
$p \wedge p = p$	Idempotent Law
$p = p$	Idempotent Law
$p \wedge TRUE = p$	Properties of TRUE and FALSE
$p \vee TRUE = TRUE$	Properties of TRUE and FALSE

$p \vee FALSE = p$	Properties of TRUE and FALSE
$p \wedge FALSE = FALSE$	Properties of TRUE and FALSE
$p \vee p = p$	Identity Law
$p \wedge p = p$	Identity Law
$p \rightarrow q = \neg p \vee q$	Law of Implication
$p \wedge (p \vee q) = p$	Absorption Law
$p \vee (p \wedge q) = p$	Absorption Law
$\neg(p \vee q) = \neg p \wedge \neg q$	De Morgan's

Jadual 1: Hukum Mantik
Table 1: The Laws of Logic

4. a) Biar

Let

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

i) Dapatkan

Find A^2

ii) Tunjukkan bahawa

Show that $A^2 - 4A + 3I = 0$

iii) Seterusnya atau sebaliknya, dapatkan

Hence, or, otherwise find A^{-1}.

(10 markah/*marks*)

b) Diberi matriks:

Given the matrix:

$$B = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix}$$

Dapatkan

Find

i) Matriks kofaktor untuk B

The cofactor matrix of B

ii) Adjoin untuk B

The ad-joint of B

iii) Matriks songsang, B^{-1}

The inverse matrix, B^{-1}

(10 markah/*marks*)

5. a) i) Berapa banyak carakah terdapat untuk menyusun huruf-huruf perkataan "ALGORITHMS"?

How many ways are there to arrange the letters of the word ALGORITHMS?

ii) Berapa banyak cara berbezakah terdapat untuk menyusun huruf-huruf dari "ALGORITHMS"sekiranya huruf vokal ditempatkan bersama?

In how many different ways can the letters of the word ' ALGORITHMS ' be arranged such that the vowels must always come together?

iii) Berapa banyak susunankah yang menggunakan huruf-huruf dari "ALGORITHMS" mempunyai **G** sebagai abjad pertama?

*How many of the arrangements made with the letters of the word ALGORITHMS have **G** as the first letter?*

(10 markah/*marks*)

b) i) Dapatkan *r* sekiranya / *Find r if*

$P(n, r) = 6\ C(n, r)$

ii) Dapatkan *n* sekiranya / Find *n* if

P (n, n) = 5040

iii) Tunjukkan bahawa /Show that

$P(n+1, r)\ (n+1-r) = P\ (n, r)\ (n+1)$

(10 markah/*marks*)

6. a) Lakarkan satu graf berarah, sekiranya boleh, sepadan dengan penerangan berikut. Bagi kes di mana graf tidak boleh dilakar, nyatakan sebab-sebabnya.

Draw a directed graph, if possible, corresponding to the following descriptions. For cases which are impossible to draw, give your reason.

i) 4 bucu, 5 darjah masuk 6 darjah keluar.

4 vertices, 5 in-degree, 6 out-degree.

ii) 4 bucu dengan 3 nod bernombor ganjil.

4 vertices with 3 odd degree nodes.

iii) 4 bucu dengan 2 nod bernombor ganjil.

4 vertices with 2 odd degree nodes.

(10 markah/*marks*)

b) Biar $G_1 = (V, E)$ merupakan satu graf tidak berarah seperti yang diberi dalam Rajah 1 berikut.

Let $G_1 = (V, E)$ be an undirected graph as shown in the Figure 1.

i) Senaraikan semua laluan dari A ke G_1.

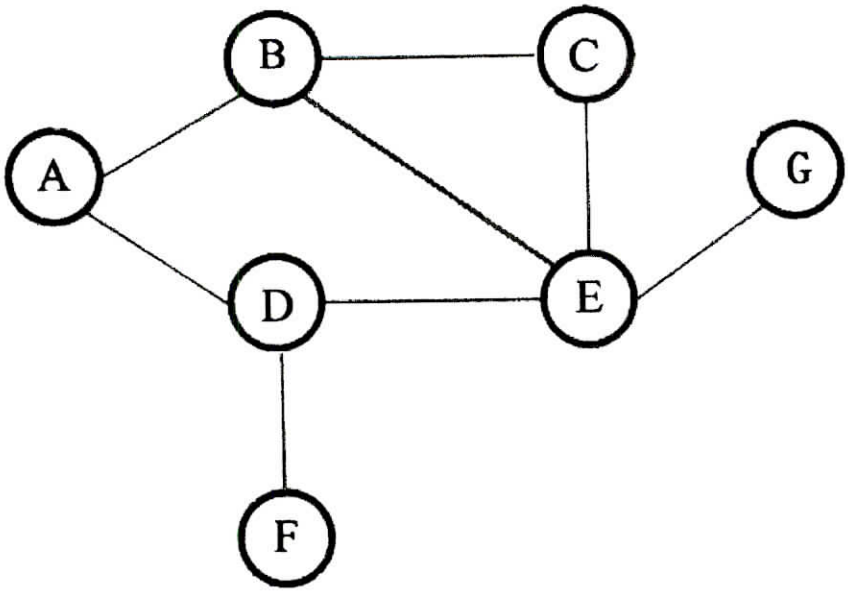
List all the paths from A to G_1.

ii) Apakah laluan yang terpanjang?

What is the length of the longest path?

iii) Bina satu matriks "adjacency" untuk graf G_1.

Build the adjacency matrix for graph G_1.



Rajah 1: $G_1 = (V, E)$
Figure 1: $G_1 = (V, E)$

(10 markah/*marks*)

TAMAT
END

UNIVERSITI MALAYA
UNIVERSITY OF MALAYA

PEPERIKSAAN IJAZAH SARJANA MUDA SAINS KOMPUTER / SARJANA MUDA TEKNOLOGI MAKLUMAT
EXAMINATION FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE / BACHELOR OF INFORMATION TECHNOLOGY

SESI AKADEMIK 2017/2018 : SEMESTER I
ACADEMIC SESSION 2017/2018 *SEMESTER I*

WIX1001: Matematik Pengkomputeran I
Computing Mathematics I

Jan 2018
Jan 2018

Masa: 2:00 jam
Time: 2:00 hours

ARAHAN KEPADA CALON:
INSTRUCTIONS TO CANDIDATES:

Calon dikehendaki menjawab **LIMA (5)** soalan sahaja.
*Answer **FIVE (5)** questions only.*

(Kertas soalan ini mengandungi 7 soalan dalam 6 halaman bercetak)
(This question paper consists of 7 questions on 6 printed pages)

1. Satu tinjauan telah dilakukan di kalangan siswa Tahun I untuk menentukan faktor yang mempengaruhi keputusan pelajar untuk melanjutkan pelajaran ke Universiti Malaya (UM). Tinjaun ini berdasarkan 3 faktor utama, iaitu, Tekanan Rakan Sebaya, Pengaruh Ibu-bapa dan Reputasi UM.
A survey was conducted on a group of First-Year students to determine the factors that influenced students' decisions to choose to attend the University of Malaya (UM). The survey was based on three main factors, i.e. Peer Pressure, Parental Influence and UM's Reputation.

Satu analisis laporan tinjauan ini mendedahkan maklumat berikut:
An analysis of the response to the survey revealed the following information:

Faktor ***Factors***	**Bilangan** ***Numbers***
Tekanan Rakan Sebaya *Peer Pressure*	56
Pengaruh Ibu Bapa *Parental Influence*	54
Reputasi UM *UM's Reputation*	64
Tekanan Rakan Sebaya dan Pengaruh Ibu Bapa tetapi bukan Reputasi UM *Peer Pressure and Parental Influence but not UM's Reputation*	34
Pengaruh Ibu Bapa dan Reputasi UM *Parental Influence and UM's Reputation*	36
Tekanan Rakan Sebaya dan Reputasi UM *Peer Pressure and UM's Reputation*	36
Tekanan Rakan Sebaya, Pengaruh Ibu Bapa dan Reputasi UM *Peer Pressure, Parental Influence and UM's Reputation*	24
Tiada Faktor yang di atas *None of the above Factors*	5

i) Berapa ramaikah orang yang mengambil bahagian dalam tinjauan ini?
How many people took part in the survey?

ii) Berapa ramaikah orang yang dipengaruhi oleh Reputasi UM tetapi tidak oleh ibu bapa atau rakan–rakan?
How many people were influenced by UM's Reputation, but not by their parents or peers?

iii) Berapa ramaikah di antara mereka yang diyakini oleh ibu bapa atau Reputasi UM?
How many were convinced by their parents or UM's Reputation?

iv) Berapa ramaikah di antara mereka yang memberi hanya satu sebab?
How many gave exactly one reason?

v) Berapa ramaikah di antara mereka yang memberi hanya dua sebab?
How many gave exactly two reasons?

(10 markah/*marks*)

2. Pertimbangkan hubungan berikut yang ditakrif atas set W yang terdiri daripada perkataan-perkataan dalam kamus Bahasa Inggeris.

Consider the following relations defined on the set W , which denotes the words in the English dictionary.

$R = \{(x, y) \in W \times W$, di mana perkataan x dan y mempunyai sekurang-kurangnya 1 abjad sepunya$\}$

R
$= \{(x, y) \in W \times W, where\ the\ words\ x\ and\ y\ have\ at\ least\ one\ letter\ in\ common\}$

Dengan mengunakan alasan, tentukan jika hubungan R:
Stating your reasons, determine if relation R *is*:

i) Refleksif
Reflexive

ii) Simetrik
Symmetric

iii) Transitif
Transitive

iv) Setara
Equivalent

(10 markah/*marks*)

3. Tentukan samada fungsi-fungsi yang diberi berikut adalah bijektif. Nyatakan sebab anda.
Determine if the following functions are bijective. State your reasons.

i) $f: R \rightarrow R,\ f(x) = 3x - 2$

ii) $g(x) = \frac{1}{1-x}, x \in (-\infty, 1)$

iii) $h: N \rightarrow N,\ h(x) = 5 - x - x^2$

(10 markah/*marks*)

4. i) Terdapat 8 orang lelaki dan 7 orang perempuan yang sanggup menganggotai satu jawatankuasa. Berapa jawatan kuasa yang terdiri daripada 8 ahli dapat dibentuk sekiranya satu jawatankuasa terdiri daripada:
There are 8 boys and 7 girls willing to join a committee. How many 8-member committees are possible if a committee is to contain:

a) Mana–mana jumlah lelaki atau perempuan?
Any number of boys and girls?

b) Lelaki sahaja?
Only boys?

c) Sekurang-kurangnya satu ahli dari setiap jantina?
At least one member of each gender?

ii) Buktikan bahawa:
Proof that:

$C(n + r, n - r) = C(n + r, 2r)$

a) sekiranya
if

b) $P(n,r) = n,$

hanya terdapat 1 nilai yang mungkin bagi r
there is only 1 possible value for r.

(10 markah/*marks*)

5. i) Berikut adalah sistem persamaan linear dengan parameter s.
The following is a system of linear equations with a parameter s.

$$\begin{cases} 3sx - 2y = 4 \\ -6x + sy = 1 \end{cases}$$

a) Tentukan nilai x and y dari segi s dengan mengguna kaedah Cramer.
Determine the value of x and y in term of s using Cramer method.

b) Sekiranya sistem persamaan ini mempunyai penyelesaian unik, iaitu, terdapat satu nilai unik bagi y untuk x yang diberi, tentukan julat s.
If the system of equations has unique solution, i.e. there is a unique value of y for an x given, determine the range of s.

ii) Diberi /*Given that* $A = \begin{pmatrix} 5 & 3 & -7 \\ 0 & -3 & 2 \\ 2 & -2 & 2 \end{pmatrix}$ dan/*and* $B = \begin{pmatrix} 9 & -3 & 8 \\ 2 & -1 & -1 \\ 5 & 0 & 5 \end{pmatrix}$.

Dapatkan/*Find*

a) $2A - B$

b) Transpos/*Transpose of* A

c) B^2

d) $\text{Det}(A) + \text{Det}(B)$

(10 markah/*marks*)

Peraturan Mantik
The Laws of Logic

$p \vee \neg\neg p = T$	Negation Law
$p \wedge \neg\neg p = F$	Negation Law
$\neg\neg p = p$	Double Negation
$p \wedge TRUE = p$	Identity
$p \vee FALSE = p$	Identity
$p \vee TRUE = TRUE$	Domination Law
$p \wedge FALSE = FALSE$	Domination Law
$p \vee p = p$	Idempotent Law
$p \wedge p = p$	Idempotent Law
$p = p$	Idempotent Law
$p \vee q = q \vee p$	Commutative Law
$p \wedge q = q \wedge p$	Commutative Law
$(p \wedge q) \wedge r = p \wedge (q \wedge r)$	Associative Law
$(p \vee q) \vee r = p \vee (q \vee r)$	Associative Law
$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	Distributive Law
$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$	Distributive Law
$p \rightarrow q = \neg p \vee q$	Law of Implication
$p \wedge (p \vee q) = p$	Absorption Law
$p \vee (p \wedge q) = p$	Absorption Law
$\neg(p \vee q) = \neg p \wedge \neg q$	De Morgan's

6. Tentukan samada proposisi berikut adalah setara dengan menggunakan Peraturan Mantik.
Determine if the following propositions are equivalent using the Laws of Logic.

i) $\neg\{(\neg p \wedge \neg q) \rightarrow (\neg p \vee \neg q)\} = F$

ii) $\neg(p \wedge \neg(\neg p \wedge q)) \equiv \neg p$

iii) $(p \vee \neg q) \vee ((\neg p \wedge \neg r) \wedge q) \equiv (p \vee \neg r) \vee \neg q$

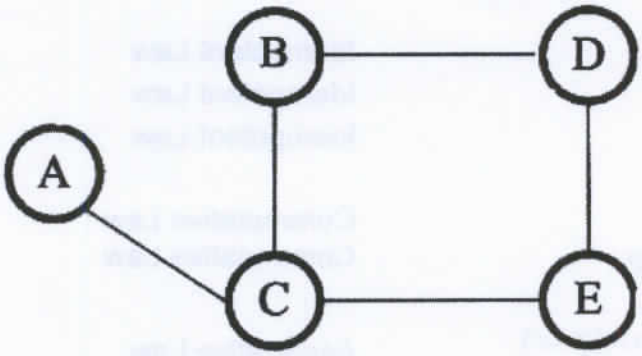
(10 markah/*marks*)

7. Biar G = (V, E) satu graf tidak berarah seperti yang ditunjukkan dalam Rajah 1.
Let G = (V, E) be the undirected graph as shown in Figure 1.

   i) Dapatkan darjah bagi setiap bucu untuk graf G.
   Find the degree of each vertex of Graph G.

   ii) Binakan matriks insiden untuk G.
   Build the incidence matrix for G.

   iii) Dapatkan senarai "adjacency" bagi setiap bucu.
   Give the adjacency lists of each vertex.



Rajah 1: Graf G
Figure 1 Graph G

(10 markah/*marks*)

TAMAT
END

UNIVERSITI MALAYA
UNIVERSITY OF MALAYA

PEPERIKSAAN IJAZAH SARJANA MUDA SAINS KOMPUTER / SARJANA MUDA TEKNOLOGI MAKLUMAT
EXAMINATION FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE / BACHELOR OF INFORMATION TECHNOLOGY

SESI AKADEMIK 2018/2019 : SEMESTER I
ACADEMIC SESSION 2018/2019 : SEMESTER I

WIX1001 : Matematik Pengkomputeran I
Computing Mathematics I

Jan 2019
Jan 2019

Masa: 2 jam
Time: 2 hours

ARAHAN KEPADA CALON :
INSTRUCTIONS TO CANDIDATES :

Calon dikehendaki menjawab **SEMUA** soalan (50 markah).
*Answer **ALL** questions (50 marks).*

(Kertas soalan ini mengandungi 5 soalan dalam 4 halaman yang dicetak)
(This question paper consists of 5 questions on 4 printed pages)

1. Sila jawab soalan-soalan yang berikut:

   Please answer the following questions:

   a) Selesaikan untuk $x : x^2 = 24 - 10x,\ x \in W$

      Solve for $x : x^2 = 24 - 10x,\ x \in W$

   b) Diberi $\mathrm{X} = \{2, 4, 5, 6\}$ dan $\mathrm{Y} = \{3, 4, 7, 8\}$, carikan yang berikut:

      Let $\mathrm{X} = \{2, 4, 5, 6\}$ and $\mathrm{Y} = \{3, 4, 7, 8\}$, *find the following:*

      $$(\mathbf{X} - \mathbf{Y})\ \mathbf{U}\ (\mathbf{Y} - \mathbf{X})$$

   c) Setiap pelajar di dalam sebuah kelas dengan 40 orang pelajar bermain sama ada permainan *Mobile Legend*, *DOTA* atau *CS GO*. 18 pelajar bermain *Mobile Legend*, 20 bermain *CS GO*, dan 27 bermain *DOTA*. 7 pelajar bermain *Mobile Legend* dan *CS GO*, 12 bermain *CS GO* dan *DOTA*, dan 4 bermain *Mobile Legend*, *DOTA* dan *CS GO*. Hitung bilangan pelajar yang bermain

      1. *Mobile Legend* dan *DOTA*
      2. *Mobile Legend* dan *DOTA* tetapi tidak bermain *CS GO*

      Each student in a class of 40 plays either Mobile Legend, DOTA or CS GO. 18 play Mobile Legend, 20 play CS GO, and 27 play DOTA. 7 play Mobile Legend and CS GO, 12 play CS GO and DOTA, and 4 play Mobile Legend, DOTA and CS GO. Find the number of students who play

      3. *Mobile Legend and DOTA*
      4. *Mobile Legend and DOTA but not CS GO.*

      (10 markah/*marks*)

2. Sila jawab soalan-soalan yang berikut:

   Please answer the following questions

   a) Adakah hubungan berikut adalah fungsi? Berikan justifikasi kepada jawapan anda. Jika ia adalah fungsi, tentukan sama ada ianya adalah fungsi satu-kepada-satu atau tidak.

      Is the following relation a function? Justify your answer. If it is a function, determine whether it is a one-to-one function or not.

      a. $R_1 = \left\{(2, 3), \left(\frac{1}{2}, 0\right), (2, 7), (-4, 6)\right\}$
      b. $R_2 = \{(x, |x|) : x \in R\}$

b) Satu hubungan R, ditakrifkan ke atas satu set integer positif sebagai aRb sekiranya "a/b", yang bermaksud a adalah faktor ke atas b. Tunjukkan dan jelaskan sama ada hubungan R adalah:

A relation R is defined on the set of positive integers as aRb if "a/b", which means that a is a factor of b. Show and explain whether the relation R is:
(1) Refleksif/ *Reflexive*
(2) Simetrik/ *Symmetric*
(3) Transitif/ *Transitive*
(4) Setara/ *Equivalence*

(10 markah/*marks*)

3. Sila jawab soalan-soalan yang berikut beserta dengan pecahannya
Please answer the following questions along with its sub-questions

a) Menggunakan jadual kebenaran, tentukan sama ada $(p \vee \sim p) \wedge (q \vee r)$ dan $(q \vee r)$ adalah setara secara logik

Using truth tables, determine if statements $(p \vee \sim p) \wedge (q \vee r)$ *and* $(q \vee r)$ *are logically equivalent*

b) Diberi P sebagai satu pernyataan "Matematik adalah mudah" dan Q adalah satu pernyataan "Saya tidak perlu belajar". Tuliskan dalam perkataan, pernyataan-pernyataan yang sepatutnya bagi yang berikut, dan ringkaskan sekiranya boleh:

Let P be the statement "Mathematics is easy" and Q be the statement "I do not need to study". Write down in words the following statements, and simplify if possible:

(1) $P \vee Q$
(2) $P \wedge Q$
(3) $\sim \sim Q$
(4) $\sim P \wedge Q$
(5) $P \Rightarrow Q$

(10 markah/*marks*)

4. Sila jawab soalan-soalan yang berikut beserta dengan pecahannya
Please answer the following questions along with its sub-questions

a) Selesaikan sistem persamaan yang berikut menggunakan kaedah *Cramer*
Solve the following system of equations using Cramer's Method

$$\begin{cases} x + 2y - z = 7 \\ 2x - 3y - 4z = -3 \\ x + y + z = 0 \end{cases}$$

b) Satu kertas soalan mengandungin bahagian P dan Q. Setiap bahagian mengandungi 10 soalan. Sekiranya seorang pelajar perlu memilih 8 soalan daripada bahagian P, dan 4 daripada bahagian Q, berapakah bilangan cara pemilihan soalan yang boleh beliau lakukan?

A question paper has two parts P and Q, each containing 10 questions. If a student needs to choose 8 from part P and 4 from part Q, in how many ways can he do that?

(10 markah/*marks*)

5. Lukis graf dengan ciri-ciri yang diberikan atau terangkan sebab mengapa graf tersebut tidak wujud

Either draw a graph with the following specified properties, or explain why no such graph exists:

a) Satu graf dengan empat *vertices*, dengan darjah setiap *vertices* adalah 1, 2, 3 dan 4.

A graph with four vertices having the degrees of its vertices 1, 2, 3 and 4.

b) Satu graf ringkas dengan lima *vertices*, dengan darjah setiap *vertices* adalah 2, 3, 3, 3 dan 5.

A simple graph with five vertices with degrees 2, 3, 3, 3, and 5.

c) Satu graf ringkas di mana setiap *vertex* berdarjah 3 dan mempunyai 6 *edges*.

A simple graph in which each vertex has degree 3 and which has exactly 6 edges.

d) Satu graf dengan empat vertices dengan vertices berdarjah 1, 1, 2 dan 6.

A graph with four vertices having the degrees of its vertices 1, 1, 2 and 6.

(10 markah/*marks*)

TAMAT
END

UNIVERSITI MALAYA
UNIVERSITY OF MALAYA

PEPERIKSAAN IJAZAH SARJANA MUDA SAINS KOMPUTER / SARJANA MUDA TEKNOLOGI MAKLUMAT
EXAMINATION FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE / BACHELOR OF INFORMATION TECHNOLOGY

SESI AKADEMIK 2019/2020 : SEMESTER I
ACADEMIC SESSION 2019/2020 : SEMESTER I

WIX1001 : Matematik Pengkomputeran I
Computing Mathematics I

Dis 2019/Jan 2020
Dec 2019/Jan 2020

Masa: 2 jam
Time: 2 hours

ARAHAN KEPADA CALON:
INSTRUCTIONS TO CANDIDATES:

Calon dikehendaki menjawab **SEMUA** soalan (50 markah).
*Answer **ALL** questions (50 marks).*

Penggunaan kalkulator dibenarkan dalam peperiksaan ini.
The use of calculators are permitted in this examination.

(Kertas soalan ini mengandungi 5 soalan dalam 6 halaman yang dicetak)
(This question paper consists of 5 questions on 6 printed pages)

1.

a) Diberi $M = \begin{bmatrix} 4 & 2 & 1 \\ 5 & 3 & 4 \\ 1 & 1 & 2 \end{bmatrix}$. Cari:

Given $M = \begin{bmatrix} 4 & 2 & 1 \\ 5 & 3 & 4 \\ 1 & 1 & 2 \end{bmatrix}$. *Find:*

i) $2M$
ii) M^2
iii) $3M^2 - 2M$

(3 markah/*marks*)

b) Merujuk kepada matriks berikut:

Refer to the following matrix:

$$A = \begin{pmatrix} 1 & -3 & 7 \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{pmatrix}$$

i) Dengan tidak menggunakan kalkulator, cari penentu A dan terangkan bagaimana anda boleh mengenalpasti bahawa A adalah matriks boleh songsang.

Without using a calculator, find the determinant of A and explain how you can identify A is an invertible matrix.

(3 markah/*marks*)

ii) Cari minors, cofactors dan adjoint matriks tersebut. Dengan itu, cari matriks songsangannya.

Find the minors, cofactors and adjoint of the matrix. Hence, find the inverse of the matrix.

(4 markah/*marks*)

2. a) Rujuk kepada hipotesis berikut:

"Mary Poppins akan membuang payungnya kalau dia tahu Mickey bercuti bersama dengan Woody tetapi Toothless mahu terbang bersama Hiccup. Toothless tidak sabar untuk terbang bersama Hiccup dan Mary Poppins tidak akan membuang payungnya."

Cadangan-cadangan dalam hipotesis tersebut telah diberi simbol seperti berikut:

P : Mary Poppins akan membuang payungnya
Q : Mickey bercuti bersama dengan Woody
R : Toothless mahu terbang bersama Hiccup

Refer to the following hypothesis:

"Mary Poppins will throw her umbrella if she knows that Mickey have his holiday with Woody but Toothless wants to fly with Hiccup. Toothless is looking forward to fly with Hiccup and Mary Poppins is not going to throw her umbrella."

The propositions in the hypothesis have been symbolized as follows:

P : Mary Poppins will throw her umbrella
Q : Mickey does have his holiday with Woody
R : Toothless wants to fly with Hiccup

i) Tulis semula hipotesis tersebut dalam bentuk simbol.

Rewrite the hypothesis in symbolic form.

(3 markah/*marks*)

ii) Dengan menggunakan peraturan kesimpulan, tentukan sama ada kesimpulan "Mickey tidak bercuti dengan Woody" adalah benar atau tidak.

Using rule of inferences, determine whether the conclusion "Mickey does not have his holiday with Woody" is valid or not.

(3 markah/*marks*)

b) Diberi kedua-dua x dan y adalah nombor nyata, kita menakrifkan:
$P(x,y)$: x boleh dibahagi oleh y
$Q(x,y)$: $\frac{x}{y} = 1$
Tentukan nilai benar setiap yang berikut:

Given that both x and y are real numbers, we define:
P(x,y) : x is divisible by y

$Q(x,y) : \frac{x}{y} = 1$

Determine the truth value of each of the following:

i) $\forall x \forall y\ Q(x,y)$
ii) $\forall x \exists y\ P(x,y)$
iii) $\forall x \forall y\ (\ Q(x,y) \not\rightarrow P(x,y)\ )$
iv) $\forall x \forall y\ (\ P(x,y) \rightarrow Q(x,y)\ )$

(4 markah/*marks*)

3. a) Seorang pensyarah mempunyai tujuh (7) buah buku pengaturcaraan dengan tajuk berlainan di atas sebuah rak buku. Empat (4) daripada buku itu berkaitan dengan 'Java' dan yang lain berkaitan dengan 'Pascal'. Dengan berapa carakah pensyarah boleh menyusun buku-buku ini di atas rak tersebut jika:

A lecturer has seven (7) programming books with different titles on a bookshelf. Four (4) of these books are on 'Java' and the rest are on 'Pascal'. In how many ways can the lecturer arrange these books on the shelf if:

i) tiada sekatan, pensyarah boleh menyusun buku-buku itu sesuka hatinya;

there is no restriction, the lecturer can arrange the books in any way he wants;

ii) semua buku 'Pascal' mesti bersebelahan dengan yang lain;

all the 'Pascal' books must be next to each other;

iii) buku-buku itu mesti disusun mengikut tajuk mereka dalam susunan menaik, dari kiri ke kanan.

the books should be arranged according to their title in an ascending order, from left to right.

(6 markah/*marks*)

b) Tujuh orang ahli falsafah, Kant, Hegel, Marx, Weber, Simmel, Heidegger dan Nietzsche sedang makan di sebuah meja bulat. Berapa carakah mereka boleh menyusun kedudukan mereka?

Seven philosophers, Kant, Hegel, Marx, Weber, Simmel, Heidegger and Nietzsche are dining at a round table. In how many ways can these philosophers be seated?

(2 markah/*marks*)

c) Berapa carakah 9 buku yang berbeza boleh dibahagikan kepada dua orang ahli falsafah, Kant dan Hegel, jika Kant mendapat 6 buku dan Hegel mendapat 3 buku?

In how many ways can 9 distinct books be distributed to two philosophers, Kant and Hegel, if Kant gets 6 books and Hegel gets 3 books?

(2 markah/*marks*)

4. a) Diberi $A = \{1, 2, 3, 4, 5\}$, $S = \{(x,y): x \in A, y \in A\}$. Cari *ordered pairs* yang memenuhi syarat-syarat di bawah. Untuk setiap satu yang berikutan, nyatakan dan terangkan sama ada ia adalah hubungan refleksif, simetri, transitif dan kesetaraan:

Given $A = \{1, 2, 3, 4, 5\}$, $S = \{(x,y): x \in A, y \in A\}$. Find the ordered pairs which satisfy the conditions given below. For each of the following, state and explain whether it is reflexive, symmetric, transitive and equivalence relation:

i) $x + y = 5$

(3 markah/*marks*)

ii) $x + y < 5$

(3 markah/*marks*)

b) Jika $f : R \rightarrow R$ ditakrifkan sebagai $f(x) = 2x^2 - 9$, tentukan sama ada ianya:

If $f : R \rightarrow R$ is defined as $f(x) = 2x^2 - 9$, identify whether it is:

i) injektif

injective

ii) surjektif

surjective

(4 markah/*marks*)

5. Sekiranya sebuah kumpulan yang mengandungi seramai 5 orang: A, B, C, D dan E, berikut adalah pasangan bagi mereka yang mengenali antara satu sama lain:

- A dan C
- A dan D
- B dan C
- C dan D
- C dan E

Suppose that a group consist of 5 people: A, B, C, D, and E, the following pairs of people are acquainted with each other:

- *A and C*
- *A and D*
- *B and C*
- *C and D*
- *C and E*

i) Lukiskan graf *G* untuk mewakili situasi tersebut.

Draw a graph G to represent this situation.

ii) Senaraikan set vertex, dan set sempadan, menggunakan notasi set. Dalam kata lain, tunjukkan set *V* dan *E* bagi bucu dan sempadan, masing-masing dalam bentuk $G = \{V, E\}$.

List the vertex set, and the edge set, using set notation. In other words, show sets V and E for the vertices and edges, respectively, in G = {V, E}.

iii) Lukiskan sebuah matriks bersebelahan bagi *G*.

Draw an adjacency matrix for G.

(10 markah/*marks*)

TAMAT
END

UNIVERSITI MALAYA
UNIVERSITI MALAYA

PEPERIKSAAN IJAZAH SARJANA MUDA SAINS KOMPUTER
EXAMINATION FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE

SESI AKADEMIK 2023/2024 : SEMESTER I
ACADEMIC SESSION 2023/2024 : SEMESTER I

WIX1001 : *MATEMATIK PENGKOMPUTERAN I*
COMPUTING MATHEMATICS I

Jan / Feb 2024
Jan / Feb 2024

Masa: 2 jam
Time: 2 hour

ARAHAN KEPADA CALON:
INSTRUCTIONS TO CANDIDATES:

Calon dikehendaki menjawab **SEMUA** soalan (50 markah).
*Candidate is required to answer **ALL** questions (50 marks).*

Kalkulator **DIBENARKAN**.
*Calculator is **ALLOWED**.*

ILMU PUNCHA KEMAJUAN

(Kertas soalan ini mengandungi 4 Soalan dalam 8 halaman yang dicetak)
(This question paper consists of 4 questions on 8 printed pages)

Soalan/*Question I*

a) Tunjukkan jadual kebenaran untuk semua input yang mungkin diberikan logik di bawah. Tentukan sama ada ia tautologi, kontingensi atau percanggahan.

Show the truth table for all possible inputs given the logic below. Determine whether it is a tautology, contingency or contradiction.

i. $\big((p \rightarrow q) \wedge (\neg q)\big) \rightarrow p$

ii. $\big(\neg q \wedge (p \leftrightarrow q)\big) \rightarrow \neg(p \wedge q)$

(5 markah/*marks*)

b) Apakah domain/set di mana pernyataan di bawah adalah benar? Bolehkah domain itu menjadi nombor Imaginasi?

What is the domain/set where the statement below is true? Can the domain be Imaginary numbers?

$\forall x \forall y$: x = y

(3 markah/*marks*)

c) Memandangkan x,y $\in$ Orang di dunia di mana K(x,y) : x kenal y, kita boleh mewakili "Seseorang kenal seseorang", "Semua orang kenal semua orang", "Semua orang kenal seseorang" dan "Seseorang kenal semua orang secara simbolik dan cipta 'jika semua orang mengenali seseorang maka seseorang itu mengenali semua orang'.

Tentukan nilai kebenaran bagi proposisi $\forall x \forall y$ K(x,y), $\forall x \exists y$ K(x,y), $\exists x \forall y$ K(x,y),$\exists x \exists y$ K(x,y) :

Given that x,y $\in$ Person in the world where K(x,y): x knows y, we can represent "Someone knows somebody", "Everyone knows everybody", "Everyone knows somebody" and "Someone knows everybody' symbolically and create 'if everyone knows somebody then somebody knows everyone'.

Determine the truth value of the propositions $\forall x \forall y$K(x,y), $\forall x \exists y$ K(x,y), $\exists x \forall y$ K(x,y), $\exists x \exists y$K(x,y):

(2 markah/*marks*)

d) Jika x ialah elemen domain Nombor Nyata tentukan sama ada fungsi di bawah adalah sah dan ke mana ia dipetakan?

If x is an element of the Real numbers' domain determine whether the function below is valid and where does it map to?

$$f(x) = \sqrt{x}$$

(2 markah/*marks*)

Soalan/*Question II*

II. A)

a) Untuk hubungan $R = \{(x, y) | \, y = x^2 + 1, x \in \{1, 3, 5\}\}$, cari domain dan julat.

For the relation $R = \{(x, y) | \, y = x^2 + 1, x \in \{1, 3, 5\}\}$, *find the domain and range.*

(1 markah/*mark*)

b) Untuk hubungan $R = \{(x, y) | \, x^2 + y^2 = 25, \; x, y \in \mathbb{Z}\}$, senaraikan unsur-unsur *R*.

For the relation $R = \{(x, y) | \, x^2 + y^2 = 25, \; x, y \in \mathbb{Z}\}$, *list the elements of R.*

(1 markah/*mark*)

c) Pertimbangkan hubungan berikut sama ada hubungan refleksif, simetri dan/atau transitif pada $X = \{a, b, c\}$.

Consider the following relations whether they are reflexive, symmetric, and/or transitive on $X = \{a, b, c\}$.

i) $R = \{(a, a), (b, b), (c, c), (a, b), (b, a), (b, c), (c, b)\}$;

ii) $R = \{(a, a), (b, b), (c, c), (a, b), (a, c), (b, c)\}$;

iii) $R = \{(a, c), (c, b)\}$;

iv) $R = \{(a, a), (b, b), (c, c), (a, b), (b, a), (a, c), (c, a), (b, c), (c, b)\}$;

(4 markah/*marks*)

II. B)

Memandangkan matriks $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}, B = \begin{bmatrix} -1 & 1 \\ 0 & 0 \\ 1 & -1 \end{bmatrix}$, dan $C = \begin{bmatrix} -6 & -4 & -2 \\ 1 & 2 & 3 \end{bmatrix}$.
Semak sama ada perkara berikut boleh dikira. Jika ya, cari jawapannya.

Given the matrices $A = \begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}, C = \begin{bmatrix} -1 & 1 \\ 0 & 0 \\ 1 & -1 \end{bmatrix}$, *and* $D = \begin{bmatrix} -6 & -4 & -2 \\ 1 & 2 & 3 \end{bmatrix}$. *Check if the following can be computed. If yes, find the answer.*

a) $A+C^T$

b) AB

c) det(A)

d) A^{-1}

e) CA^TA

f) Tunjukkan dengan contoh mengapa pendaraban matriks tidak komutatif.

Demonstrate with an example, why matrix multiplication is not commutative.

(5 markah/*marks*)

II. C) *Selesaikan sistem persamaan berikut dengan menggunakan kaedah Cramer*

Solve the following system of equations by using Cramer's method

$$\begin{cases} u + v - w &= 6 \\ 2u - 3v + 2w &= -4 \\ -4u + 5v + w &= 9 \end{cases}$$

(2 markah/*marks*)

Soalan/Question III

Fakulti Sains Komputer & Teknologi Maklumat, Universiti Malaya menawarkan tiga jenis kursus matematik lanjutan: Matematik Pengkomputeran I, Matematik Pengkomputeran II, dan Analisis Berangka. Terdapat 120 pelajar mendaftar dalam sekurang-kurangnya satu daripada kursus ini. Pendaftaran bagi setiap kursus adalah seperti berikut:

The Faculty of Computer Science & Information Technology, Universiti Malaya offers three types of advanced mathematics courses: Computing Math I, Computing Math II, and Numerical Analysis. There are 120 students enrolled in at least one of these courses. The enrolment for each course is as follows:

- 70 pelajar telah mendaftar dalam Matematik Pengkomputeran I.

  70 students are enrolled in Computing Math I.

- 45 pelajar telah mendaftar dalam Matematik Pengkomputeran II.

  45 students are enrolled in Computing Math II.

- 60 pelajar telah mendaftar dalam Analisis Berangka.

  60 students are enrolled in Numerical Analysis.

Ia juga diketahui bahawa:

It is also known that:

- 20 pelajar telah mendaftar dalam kedua-dua Matematik Pengkomputeran I dan Matematik Pengkomputeran II.

  20 students are enrolled in both Computing Math I and Computing Math II.

- 25 pelajar telah mendaftar dalam kedua-dua Matematik Pengkomputeran I dan Analisis Berangka.

  25 students are enrolled in both Computing Math I and Numerical Analysis.

- 15 pelajar telah mendaftar dalam kedua-dua Matematik Pengkomputeran II dan Analisis Berangka.

  15 students are enrolled in both Computing Math II and Numerical Analysis.

- 10 pelajar telah mendaftar dalam ketiga-tiga kursus.

  10 students are enrolled in all three courses.

a) Memandangkan maklumat ini, pihak pentadbiran universiti berminat untuk mengetahui bilangan pelajar yang mendaftar dalam tepat satu kursus untuk memperuntukkan sumber yang sesuai. Kira bilangan pelajar yang mendaftar dalam satu kursus.

Given this information, the university's administration is interested in finding out the number of students who are enrolled in exactly one course to allocate appropriate resources. Calculate the number of students enrolled in exactly one course.

(4 markah/*marks*)

b) Daripada pelajar yang mendaftar dalam satu kursus matematik di fakulti, jabatan matematik memutuskan untuk mencipta satu siri kuliah khas di mana tiga penceramah pelajar jemputan akan membentangkan tentang cabaran dan pandangan unik dalam mempelajari satu bidang matematik. Jika setiap penceramah mestilah daripada satu daripada tiga kursus yang berbeza (Pengkomputeran Matematik I, Pengkomputeran Matematik II, Analisis Berangka) dan susunan mereka bercakap. Cari bilangan susunan pembesar suara yang berbeza.

From the students enrolled in exactly one mathematics course at the faculty, the mathematics department decides to create a special lecture series where three guest student speakers will present on the unique challenges and insights of studying a single area of mathematics. If each speaker must be from a different one of the three courses (Computing Math I, Computing Math II, Numerical Analysis), and the order in which they speak. Find the number of different arrangements of speakers.

(4 markah/*marks*)

c) Sebagai persediaan untuk perlawanan akhir, pelajar dari kumpulan yang sama dengan mereka yang mendaftar dalam satu kursus membuat keputusan untuk membentuk kumpulan belajar. Jika satu kumpulan belajar boleh terdiri daripada mana-mana 5 orang pelajar. Cari bilangan cara untuk membentuk satu kumpulan belajar daripada semua pelajar yang mengambil satu kursus sahaja.

In preparation for finals, students from the same group of those who are enrolled in exactly one course decide to form study groups. If a study group can consist of any 5 students. Find the number of ways to form one study group from all the students taking one course only.

(4 markah/*marks*)

Soalan/Question IV

IV. A)

a) Lukiskan graf dengan matriks bersebelahan berikut berkenaan dengan susunan bucu a, b, c, d.

Draw the graph with the following adjacency matrix with respect to the ordering of vertices a, b, c, d.

$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}$$

(2 markah/*marks*)

b) Kelaskan graf yang terhasil di atas dalam bahagian a dan namakannya.

Classify the above resulting graph in part a and name it.

(1 markah/*mark*)

c) Senaraikan darjah setiap bucu graf dalam bahagian a.

List the degree of each vertex of the graph in part a.

(1 markah/*mark*)

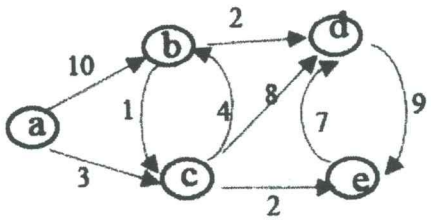
d) Bolehkah graf ringkas wujud dengan 13 bucu setiap darjah tiga? kenapa?

Can a simple graph exist with 13 vertices each of degree three? Why?

(2 markah/*mark*)

IV. B) Biarkan G = (V, E) ialah graf berwajaran seperti yang ditunjukkan dalam rajah di bawah.

Let G = (V, E) be a weighted graph as shown in the figure below.



a) Terangkan model graf.

Describe the graph model.

(2 markah/*marks*)

b) Menggunakan algoritma Dijkstra, cari panjang jarak terpendek antara bucu a dan d.

Using Dijkstra's algorithm, find the length of the shortest distance between the vertices a and d.

(1 markah/*mark*)

c) Menggunakan algoritma Dijkstra di atas, nyatakan laluan terpendek yang sepadan antara a dan d.

Using Dijkstra's algorithm above, indicate the corresponding shortest path between a and d.

(1 markah/*mark*)

IV. C) Biarkan T=(V, E) ialah pokok berakar dengan akar B yang mewakili kandungan buku. Buku ini mengandungi empat bab (C_1, C_2, C_3, C_4). Bucu dengan tahap nombor 2 ialah bahagian (S_i) dalam setiap bab, di mana setiap bab terdiri daripada 2 bahagian, masing-masing. S_4 dan S_6 tahap 2 termasuk subseksyen tunggal (E_i) dalam bahagian mereka yang mewakili tahap 3 pokok.

Let T= (V, E) be a rooted tree with root B that represents a book's contents. The book consists of four chapters (C_1 , C_2 , C_3 , C_4). Vertices with level number 2 are the sections (S_i) within each chapter, where each chapter consists of 2 sections, respectively. S_4 and S_6 of level 2 include a single subsection (E_i) within their section that represents level 3 of the tree.

a) Lukis graf pokok.

Draw the tree graph.

b) Bucu yang manakah adalah daun?

Which vertices are leaves?

c) Berapakah ketinggian pokok itu?

What is the height of the tree?

(3 markah/*marks*)

TAMAT
END

UNIVERSITI MALAYA
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PEPERIKSAAN IJAZAH SARJANA MUDA SAINS KOMPUTER
EXAMINATION FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE

SESI AKADEMIK 2024/2025 : SEMESTER 1
ACADEMIC SESSION 2024/2025 : SEMESTER 1

WIX1001 : Matematik Pengkomputeran I
Computing Mathematics I

Februari 2025 Masa : 2 jam
February 2025 Time : 2 hours

ARAHAN KEPADA CALON :
INSTRUCTIONS TO CANDIDATES :

1) Calon dikehendaki menjawab **SEMUA SOALAN** (50 markah).
*Answer **ALL QUESTIONS** (50 marks).*

2) Penggunaan kalkulator yang tidak boleh diprogramkan **DIBENARKAN** dalam peperiksaan ini.
*The use of non-programmable calculators are **ALLOWED** in this examination.*

(Kertas soalan ini mengandungi 4 soalan dalam 7 halaman yang dicetak)
(This question paper consists of 4 questions on 7 printed pages)

SOALAN 1 / *QUESTION 1*

a) Terdapat 21 ahli dalam Kelab Kejuruteraan Perisian (SE), dan 26 ahli dalam Kelab Sistem Rangkaian (NS), dan 29 ahli dalam Kelab Kepintaran Buatan (AI) FSKTM UM. Antaranya, 14 mempunyai dwi keahlian SE dan NS, 15 mempunyai dwi keahlian NS dan AI, dan 12 mempunyai dwi keahlian SE dan AI. Selain itu, 8 daripada mereka mempunyai keahlian tiga kelab AI, SE dan NS. Cari jawapan untuk unsur berikut:

There are 21 members in Software Engineering (SE) Club, and 26 members in Network Systems (NS) Club, and 29 in Artificial Intelligence (AI) Club of FCSIT UM. Among them, 14 have dual membership of SE and NS, 15 have dual membership of NS and AI, and 12 have dual membership of SE and AI. Besides, 8 of them have triple membership of AI, SE, and NS. Find the answers for the following items:

i. Tentukan jumlah ahli yang menyertai sekurang-kurangnya **SATU (1)** kelab SE, AI atau NS.

*Determine the total members who joined at least **ONE (1)** SE, AI or NS club.*

ii. Berapa ramaikah daripada mereka bukan ahli Kelab NS?

How many of them are not members of NS Club?

iii. Cari jumlah ahli dengan dua keahlian sahaja (bukan keahlian tunggal dan bukan keahlian tiga kelab).

Find total members with dual memberships only (not single memberships, and not triple memberships).

iv. Cari jumlah ahli dengan keahlian tunggal sahaja (bukan keahlian dua kelab, dan bukan keahlian tiga kelab).

Find total members with single memberships only (not dual memberships, and not triple memberships).

v. Berapa ramai daripada mereka yang mempunyai keahlian tunggal SE?

How many of them have single memberships of SE?

(5 markah / *marks*)

b) Diberi set $A = \{x : x$ ialah integer genap di antara 1 dan 10$\}$ dan set $B = \{x : x$ ialah gandaan integer 3 di antara 1 dan 10$\}$. Selesaikan perkara berikut:

Given set $A = \{x : x$ is an even integer within 1 and 10$\}$ and set $B = \{x : x$ is an integer multiple of 3 within 1 and 10$\}$. Solve the followings:

i. Cari $A \cup B$.

Find $A \cup B$.

ii. Cari $A \cap B$.

Find $A \cap B$.

(2 markah / *marks*)

c) Buktikan kesetaraan logik proposisi di bawah menggunakan hukum logik.

Proof the logical equivalence of the proposition below using laws of logic.

$$p \vee (\neg(\neg(p \wedge q)) \equiv (p \vee \neg q)$$

(3 markah / *marks*)

d) Biarkan $P(x)$ sebagai predikat "x boleh dibahagi dengan 3" di mana x ialah integer, dan $Q(y,z)$ ialah predikat "$y + z > 10$" dengan y dan z ialah nombor nyata. Tentukan nilai:

Let P(x) be the predicate "x is divisible by 3" where x is an integer, and Q(y, z) be the predicate "y + z > 10" where y and z are real numbers. Determine the truth value of:

i. $\forall x\, P(x)$

ii. $\exists x\, P(x)$

iii. $\forall x\, \exists y\, Q(x,y)$

(3 markah / *marks*)

SOALAN 2 / *QUESTION 2*

Sebuah syarikat mempunyai tiga penghala yang saling berhubung, iaitu X, Y, dan Z, yang membantu mengarahkan trafik internet. Sistem persamaan berikut boleh mewakili jumlah paket data yang dipindahkan di antara mereka setiap minit:

A company has three interconnected routers, X, Y, and Z, that help direct internet traffic. The following system of equations can represent the number of data packets transferred between them each minute:

$$\begin{cases} -y - 2z = -8 \\ x + 3z = 2 \\ 7x + y + z = 0 \end{cases}$$

di mana:

- x mewakili paket yang dihantar oleh Penghala X,
- y mewakili paket yang dihantar oleh Penghala Y, dan
- z mewakili paket yang dihantar oleh Penghala Z.

where:

- *x represents the packets routed by Router X,*
- *y represents the packets routed by Router Y, and*
- *z represents the packets routed by Router Z.*

a) Selesaikan nilai *x*, *y* dan *z* menggunakan **Kaedah Cramer** dan sahkan jawapan anda dengan **Kaedah Matriks Songsang**. Berdasarkan nilai *x*, *y* dan *z*, tentukan jika ada mana-mana penghala yang menerima paket data dan bukannya menghantarnya keluar.

*Solve the values of x, y and z using **Cramer's Method** and validate your answer with **Inverse Matrix Method**. Based on the values of x, y and z, determine if any router is receiving packets instead of routing them out.*

(10 markah / *marks*)

b) Diberikan bahawa setiap paket data mesti melalui tepat dua penghala yang berbeza secara berturut-turut (bermula dari satu penghala dan berakhir di penghala yang lain), berapakah jumlah laluan unik yang melibatkan dua penghala yang mungkin jika setiap paket boleh bermula di mana-mana penghala (X, Y, atau Z) dan mesti berakhir di penghala yang berbeza?

Given that each data packet must pass through exactly two different routers in sequence (starting from one router and ending at another), how many unique two-router routes are possible if each packet can start at any router (X, Y, or Z) and must end at a different router?

(3 markah / *marks*)

SOALAN 3 / *QUESTION 3*

a) Rujuk kepada matriks bersebelahan berikut:

Refer to the following adjacency matrix.

$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}$$

i. Lukiskan graf dengan matriks bersebelahan tersebut berkenaan dengan susunan bucu *a*, *b*, *c*, *d*.

Draw the graph with the adjacency matrix with respect to the ordering of vertices a, b, c, d.

(2 markah / *marks*)

ii. Adakah graf yang digambarkan dalam (i) mempunyai litar Euler? Terangkan jawapan anda.

Does the illustrated graph in (i) have an Euler circuit? Justify your answer.

(1 markah / *mark*)

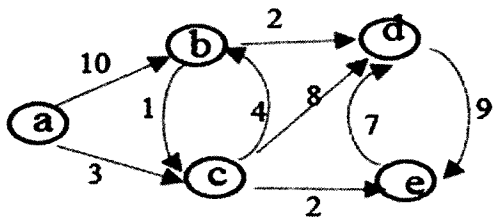
b) Bolehkah graf ringkas wujud dengan 13 bucu setiap darjah tiga? Mengapa?

Can a simple graph exist with 13 vertices each of degree three? Why?

(2 markah / *marks*)

c) Biarkan $G = (V, E)$ ialah graf berpemberat seperti yang ditunjukkan dalam rajah di bawah:

Let G = (V, E) be a weighted graph as shown in the figure below:



i. Cari *in-degree* bucu *d*.

Find the in-degree of vertex d.

(1 markah / *mark*)

ii. Menggunakan algoritma Dijkstra, cari panjang jarak terpendek antara bucu *a* dan *d*. Tunjukkan jalan kerja anda menggunakan jadual.

Using Dijkstra's algorithm, find the length of the shortest distance between the vertices a and d. Show your working table.

(3 markah / *marks*)

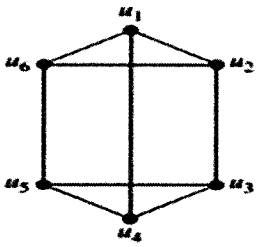
iii. Menggunakan algoritma Dijkstra, nyatakan laluan terpendek yang sepadan antara *a* dan *d*.

Using Dijkstra's algorithm, indicate the corresponding shortest path between a and d.

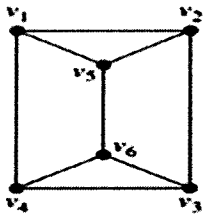
(1 markah / *mark*)

d) Tentukan sama ada pasangan graf yang diberi adalah isomorfik. Paparkan isomorfisme atau berikan sebab untuk tiadanya wujud:

Determine whether the given pair of graphs are isomorphic. Display an isomorphism or provide a reason for none exists:



H1



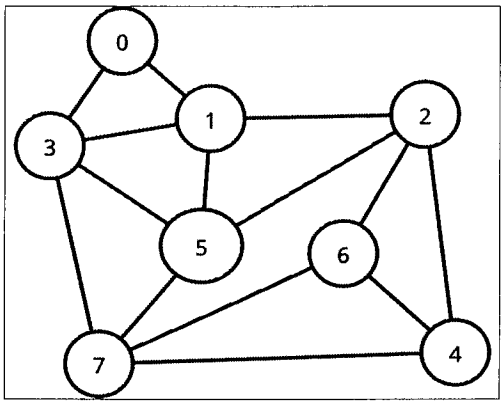
H2

(2 markah / *marks*)

SOALAN 4 / *QUESTION 4*

a) Rajah berikut menunjukkan graf dengan 8 bucu disusun dalam susunan berangka, iaitu V = (0, 1, 2, 3, 4, 5, 6, 7).

The following figure shows a graph with 8 vertices arranged in numerical order, i.e. V = (0, 1, 2, 3, 4, 5, 6, 7).



Jika *breadth-first search* digunakan untuk mencari pokok rentang bermula pada bucu 0:

If breadth-first search is used to find a spanning tree started at vertex 0:

i. Apakah 4 bucu pertama yang dilawati mengikut urutan?

What are the first 4 vertices being visited in order?

ii. Lukiskan pokok rentang ini.

Draw this spanning tree.

(4 markah / *marks*)

b) Jinnie merancang untuk membangunkan permainan komputer yang diterangkan seperti di bawah:

"Terdapat 10 stesen dalam permainan. Setiap stesen S_x, dengan x mewakili nombor asli dari 1 hingga 10, membawa 'tenaga' $E(x)$ tertentu, supaya $E(x) = \frac{1}{2}x^2 - 4x + 3$. Seorang pemain dalam permainan dibenarkan berpindah dari stesen S_x ke stesen lain S_y jika dan hanya jika perbezaan 'tenaga' tidak lebih daripada 3, iaitu $|E(x) - E(y)| \leq 3$. Semasa terdapat lebih daripada satu stesen yang mungkin untuk dipindahkan, pemain sentiasa memilih stesen dengan nilai x yang lebih rendah, iaitu S_2 mempunyai keutamaan yang lebih tinggi berbanding dengan S_3, dan S_3 mempunyai keutamaan yang lebih tinggi berbanding dengan S_4, dan seterusnya."

Jinnie mahu faham adakah permainan ini direka dengan baik. Bantu Jinnie menganalisis dengan:

Jinnie plans to develop a computer game that is described below:

"There are 10 stations in the game. Each station S_x, where x represents natural numbers from 1 to 10, carries certain 'energy' E(x), such that $E(x) = \frac{1}{2}x^2 - 4x + 3$. A player in the game is allowed to move from a station S_x to another station S_y if and only if the difference in 'energy' is not more than 3, i.e. $|E(x) - E(y)| \leq 3$. While there is more than one possible station to move to, the player always choose the station with lower x value, i.e. S_2 has higher precedence compared to S_3, and S_3 has higher precedence compared to S_4, and so on."

Jinnie wants to understand is this game well designed. Help Jinnie to analyze by:

i. Menggunakan algoritma *Depth-First Search*, bangunkan pokok rentang berdasarkan stesen ini yang berakar pada S_1. Tunjukkan cara algoritma berfungsi dan lukis pokok tersebut.

Using Depth-First Search algorithm, develop a spanning tree based on these stations rooted at S_1. Show how the algorithm works and draw the tree.

ii. Kenal pasti jika terdapat stesen yang tidak bersambung.

Identify if there is any unconnected station.

(8 markah / *marks*)

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